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Experiment No. 3

Title: Preparation of Zinc Oxide (ZnO) Nanoparticles by Precipitation / Co-precipitation Method

Aim: To synthesize zinc oxide (ZnO) nanoparticles using the co-precipitation method under alkaline conditions.

Requirements (Chemicals):

Zinc precursor salt:

- Zinc nitrate hexahydrate, $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ or
- Zinc acetate dihydrate, $\text{Zn}(\text{CH}_3\text{COO})_2 \cdot 2\text{H}_2\text{O}$ or
- Zinc sulfate heptahydrate, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ or
- Zinc chloride, ZnCl_2
- Dilute alkaline solution: 1 N NaOH / 1 N KOH
- Distilled water

Apparatus

Burette, beaker / three-neck flask, pH paper, magnetic stirrer with magnetic needle, measuring cylinder, filter paper, hot air oven, weighing balance, thermometer, etc.

Theory

The **precipitation or co-precipitation technique** is a simple, cost-effective, and widely used chemical method for the synthesis of **metal oxide and mixed metal oxide**

nanoparticles. In this method, a soluble metal salt reacts with an alkaline solution to form an **insoluble metal hydroxide precipitate**, which upon heating decomposes to form the corresponding metal oxide.

Importance of Zinc Oxide (ZnO):

Zinc oxide is an n-type wide band gap semiconductor with a band gap of approximately 3.37 eV and a high exciton binding energy (~60 meV). These properties make ZnO highly attractive for optoelectronic and electronic applications. When synthesized in nanoparticle form, ZnO exhibits enhanced surface area, quantum confinement effects, and improved optical and electrical properties.

Applications of ZnO nanoparticles include:

- UV photodetectors and light-emitting devices (LEDs)
- Gas and chemical sensors
- Transparent conducting electrodes
- Photocatalysis and solar cells
- Antibacterial and biomedical applications

The co-precipitation method allows **controlled particle size, good crystallinity, and high purity**, making it suitable for laboratory-scale synthesis of ZnO nanoparticles.

Procedure:

Zinc oxide (ZnO) nanoparticles can be synthesized under alkaline conditions using different bases, **BOH**, where

B = Na⁺, K⁺, or NH₃.

1. Preparation of precursor solution

Weigh 0.01 mol of zinc precursor salt

(Zn(NO₃)₂·6H₂O / Zn(CH₃COO)₂·2H₂O / ZnSO₄·7H₂O / ZnCl₂)

and dissolve it in 100 mL of distilled water.

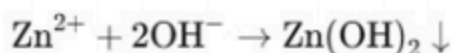
2. Transfer the solution to a 500 mL beaker or three-neck flask, and keep it in a room-temperature water bath (25–30 °C).
3. Place the solution on a magnetic stirrer and start continuous stirring.
4. Add 1 N NaOH or KOH solution drop-wise using a burette until the pH reaches 9, monitored using pH paper.



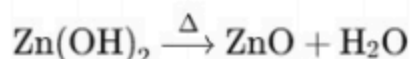
5. Continue stirring the reaction mixture for 30 minutes, during which a white precipitate of zinc hydroxide $[\text{Zn}(\text{OH})_2]$ is formed.
6. Separate the precipitate by filtration (or magnetic separation, if applicable).
7. Wash the precipitate several times with distilled water to remove excess base and by-products.
8. Dry the washed precipitate in a hot air oven at 100°C for 1 hour.
9. Upon heating, zinc hydroxide decomposes to zinc oxide nanoparticles.

Chemical Reactions Involved:

Formation of zinc hydroxide:



Conversion to zinc oxide:



[Note]

1. $\text{Zn}(\text{OH})_2$ upon heating produces ZnO
2. ZnO nanoparticles prepared by this method typically show wurtzite crystal structure and enhanced semiconducting behaviour due to nanoscale effects.

Result:

1. Weight of dried ZnO nanoparticles = _____ g